



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to P.N.ANNA PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

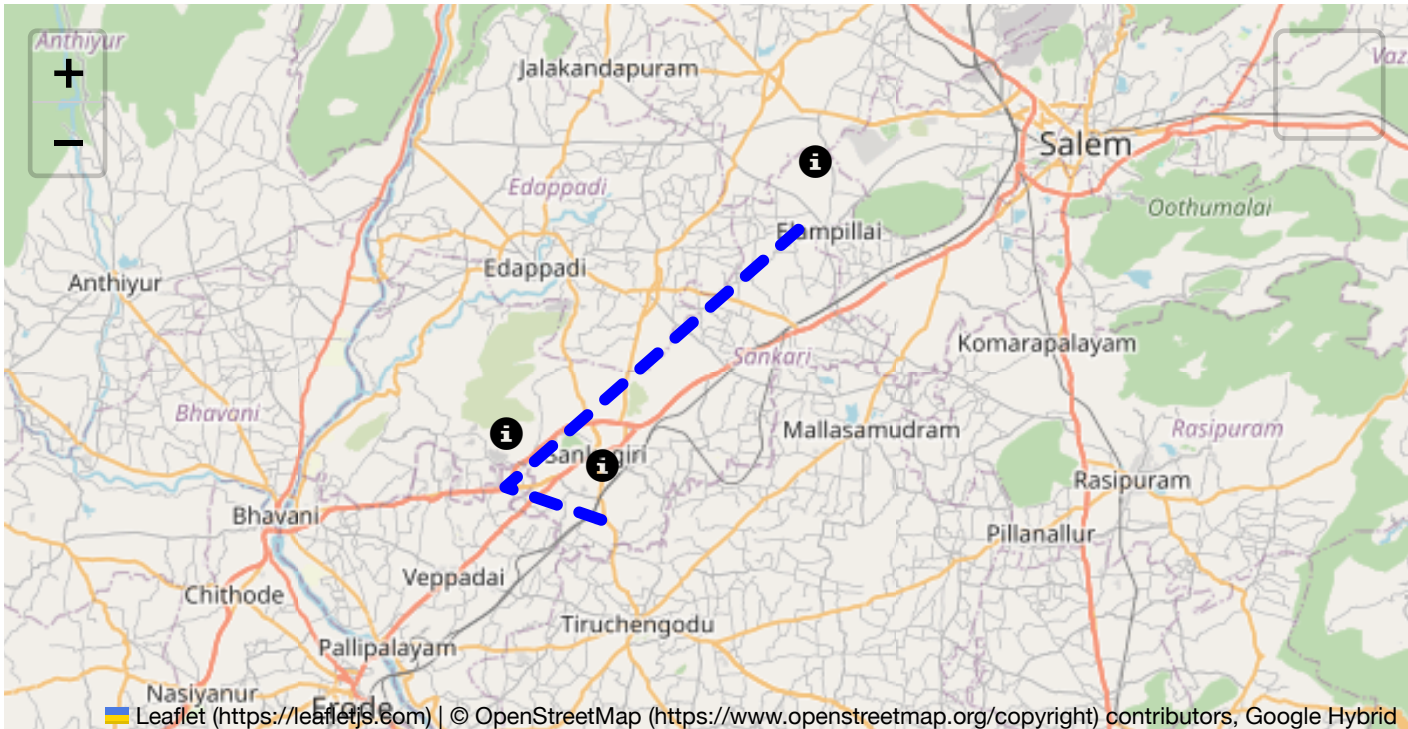
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

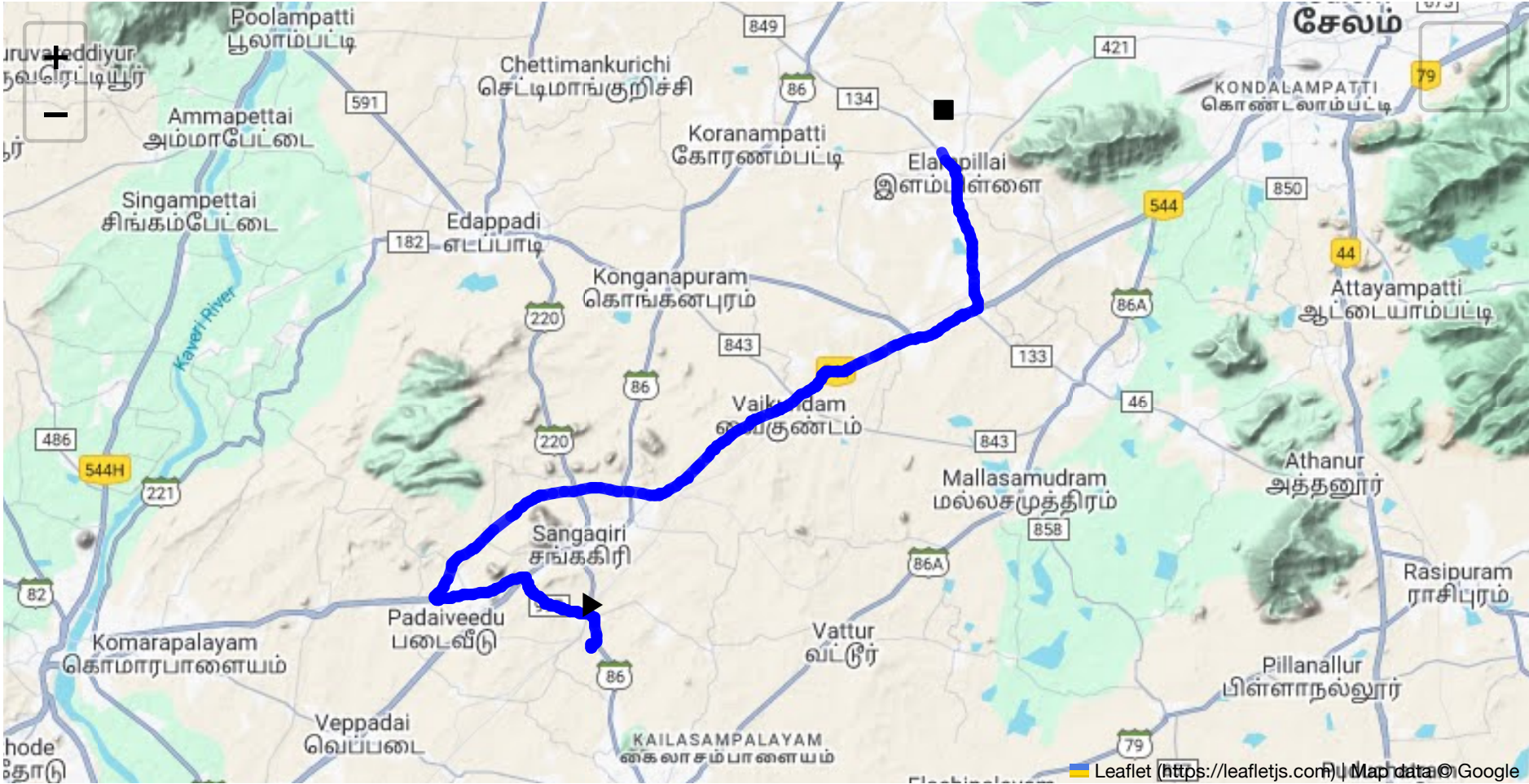
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.41 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours

Start: (11.4381, 77.8734)
End: (11.612964, 77.999176)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Elampillai via Bhavani Main Road

1. Overview of the Route Map:

- The route covers approximately 40.41 kilometers, typically requiring about 0.85 hours for heavy vehicles carrying hazardous materials.
- Starting at Sangagiri, the route follows local and state highways, with Bhavani Main Road as a significant segment, and terminates at Elampillai.
- The route traverses through semi-urban areas with a mix of rural roads.

2. Typical Weather Conditions and Potential Hazards:

- **Weather:** Tamil Nadu experiences tropical climate; hot summers, moderate monsoons (June-September), and mild winters.
- **Hazards:** Monsoon season can lead to waterlogging, reduced visibility, and slippery roads, possibly causing delays and accidents.

3. Traffic Patterns and Congestion:

- **Peak Hours:** Morning (7 AM - 10 AM) and evening (5 PM - 8 PM) see increased traffic.
- **Congestion Hotspots:** Market areas, especially near Sangagiri and Elampillai, can severely slow down traffic.
- Regular congestion along Bhavani Main Road due to its importance as a connector.

4. Road Quality and Infrastructure:

- Generally good roads, though rural segments may have narrower lanes and uneven surfaces.

- Few speed breakers and unmarked curves, which require cautious driving.
- Signage is moderate; drivers unfamiliar with the area should be vigilant.

5. Alternative Routes for Emergencies:

- **Route A:** Use the NH44 bypass to avoid congested areas near Elampillai.
- **Route B:** Via Omalur Road connecting via state highways, slightly longer but less congested.
- These alternatives offer better infrastructure for emergency maneuvers.

6. Local Regulations Affecting Hazardous Material Transport:

- Trucks transporting hazardous materials must adhere to state guidelines that often include restrictions during peak hours.
- Specific permits might be required, and there are likely checkpoints enforcing regulations.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Frequency of incidents is typically low but includes occasional rollovers during the monsoon season.
- Accidents often result from sharp turning points and unmarked rural junctions.

8. Environmental Considerations and Sensitive Areas:

- Proximity to agricultural areas; caution advised to prevent chemical spills affecting crops.
- Some segments near Sangagiri and Elampillai cross water bodies; contingency plans for spill prevention are essential.

9. Communication Coverage:

- Generally good mobile and internet coverage along state highways.
- Potential dead zones in rural areas, particularly around Padaiveedu.
- Drivers should carry physical maps and emergency communication devices.

10. Estimated Emergency Response Times:

- **Urban Segments:** Approximately 15-30 minutes.
- **Rural Segments:** Could extend to 45 minutes or more; slower response due to isolated stretches.
- Local emergency services have moderate reach; familiarize with local contacts.

12. Overall Summary of Risk Assessment:

- The route is considered moderately safe for transporting hazardous materials with proper precautions.
- Drivers should be particularly cautious during the monsoon season, around peak traffic hours, and through rural segments with patchy communication.
- Adhering to traffic regulations, continuous monitoring of weather conditions, and emergency preparedness will mitigate the majority of risks.

This comprehensive analysis serves as a safety guide for truck drivers traversing this route, emphasizing proactive measures and informed decision-making.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46307, 77.84941	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	High	11.55839, 78.01288	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium
4	hospital	SR Hospital	11.6075985, 78.0070051	30 km/h	Medium
5	hospital	Subam Hospital	11.6069532, 78.0042593	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668